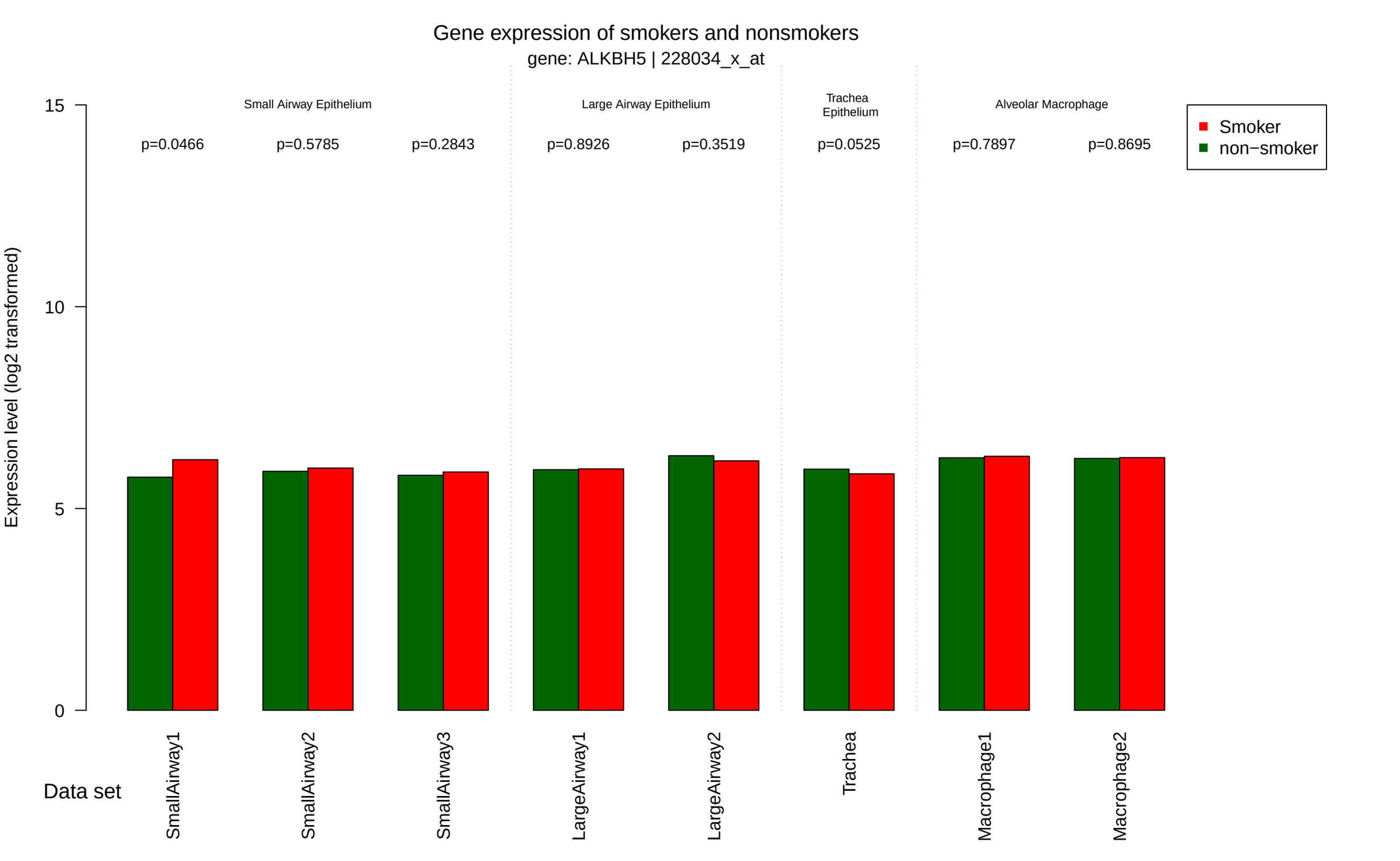
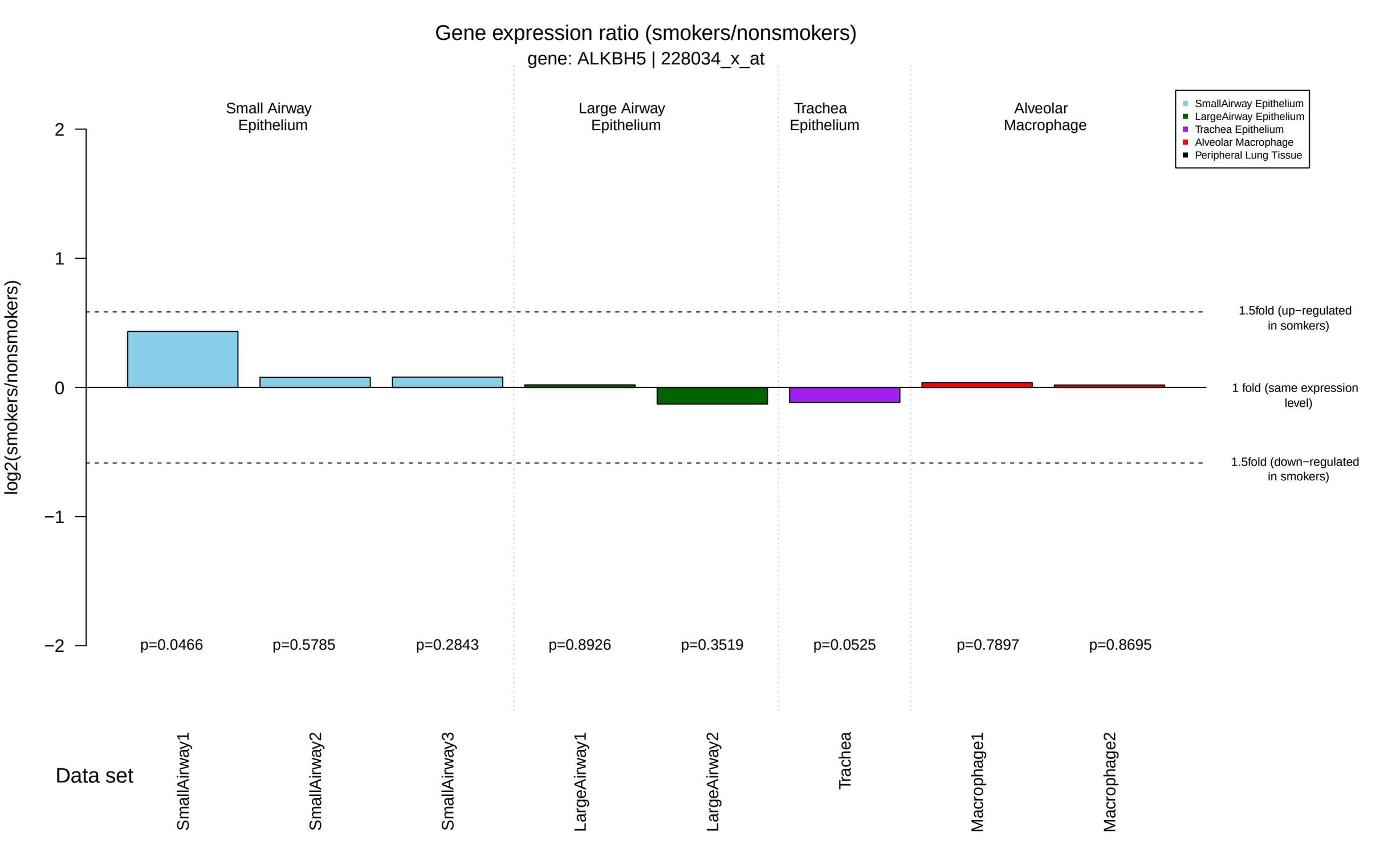
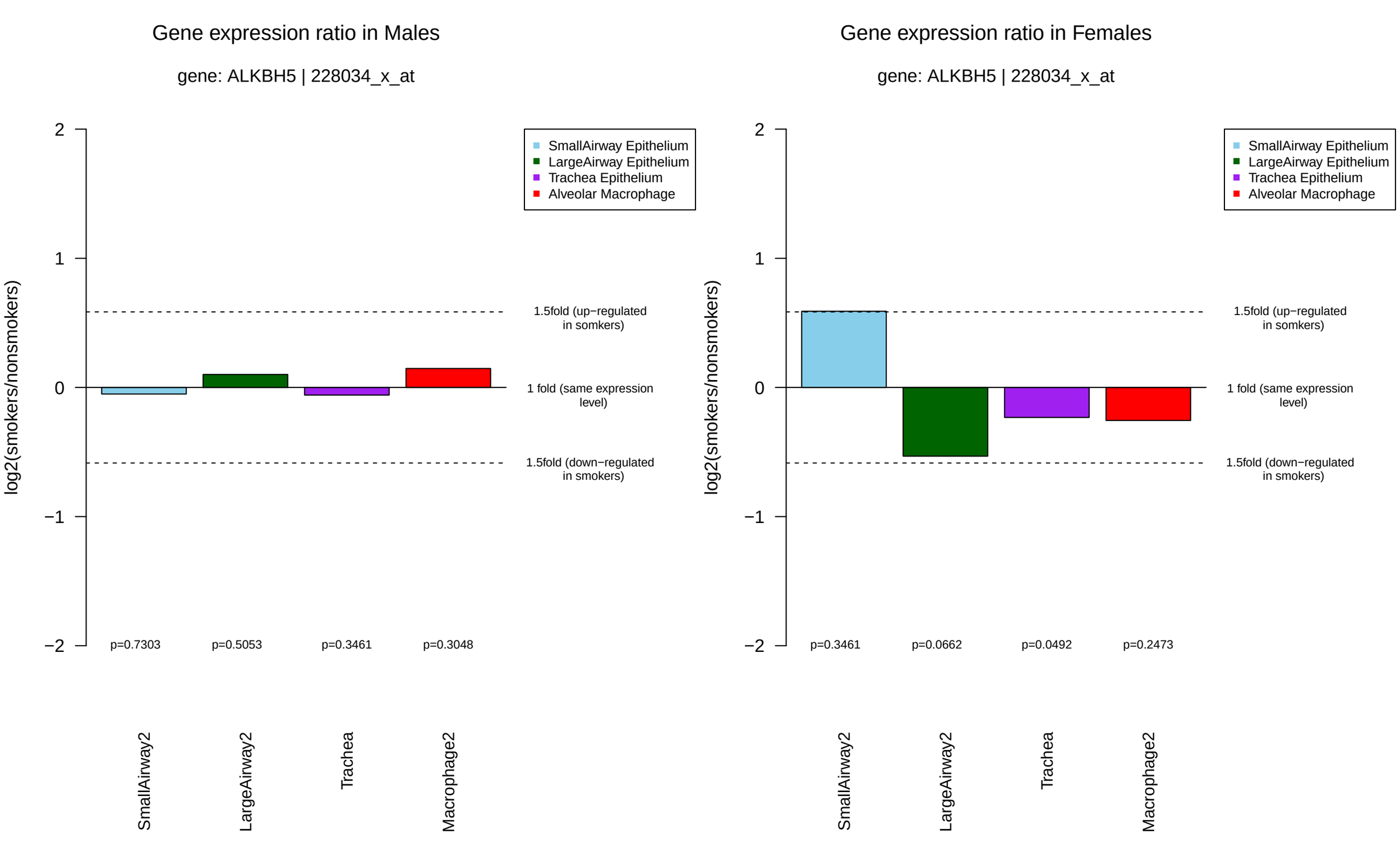
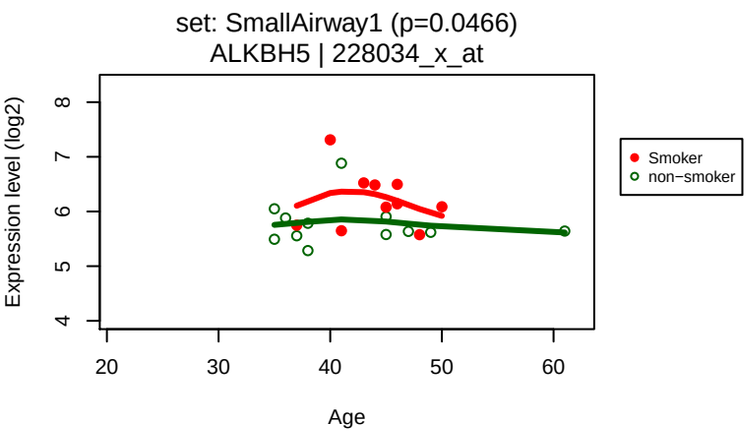
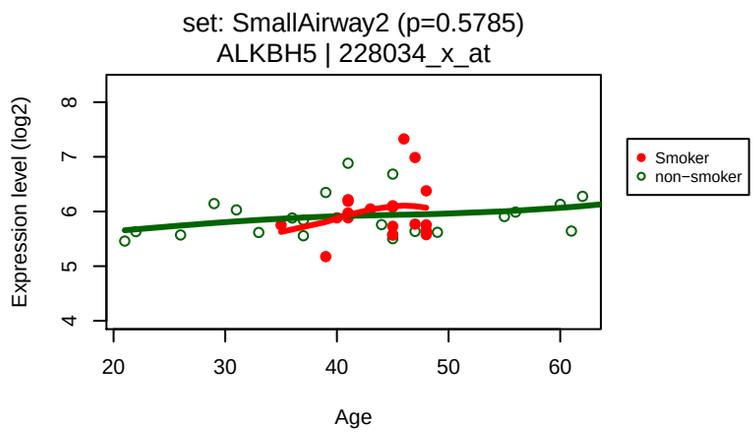
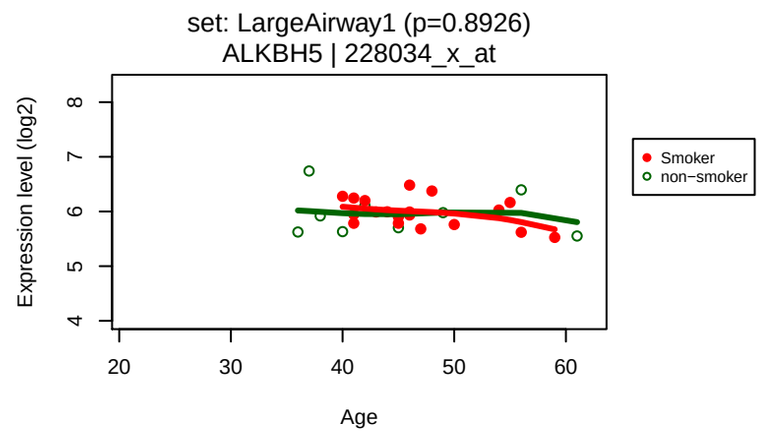

Gene expression of smokers and nonsmokers
gene: ALKBH5 | 228034_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0466
p=0.5785
p=0.2843
p=0.8926
p=0.3519
p=0.0525
p=0.7897
p=0.8695
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

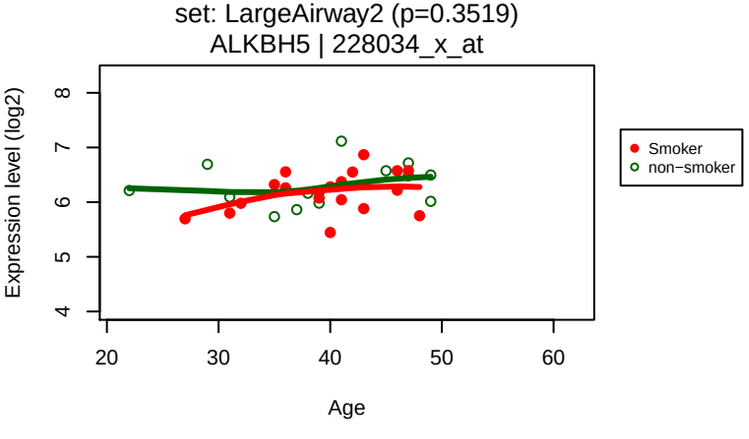

Gene expression ratio (smokers/nonsmokers)
gene: ALKBH5 | 228034_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0466
p=0.5785
p=0.2843
p=0.8926
p=0.3519
p=0.0525
p=0.7897
p=0.8695
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

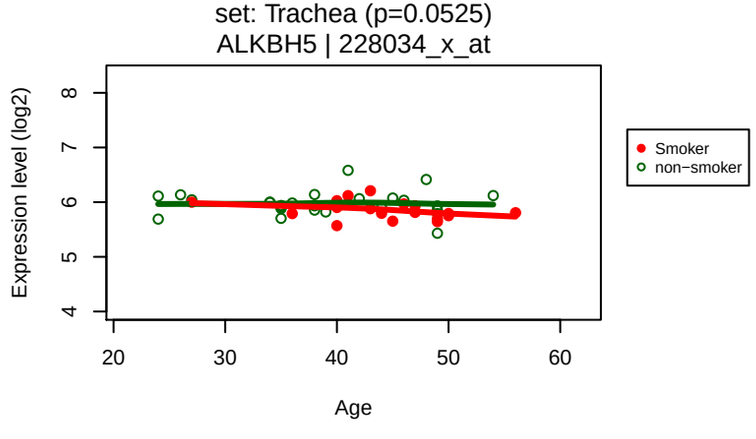

Gene expression ratio in Males
gene: ALKBH5 | 228034_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7303
p=0.5053
p=0.3461
p=0.3048
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALKBH5 | 228034_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3461
p=0.0662
p=0.0492
p=0.2473
SmallAirway2
LargeAirway2
Trachea
Macrophage2

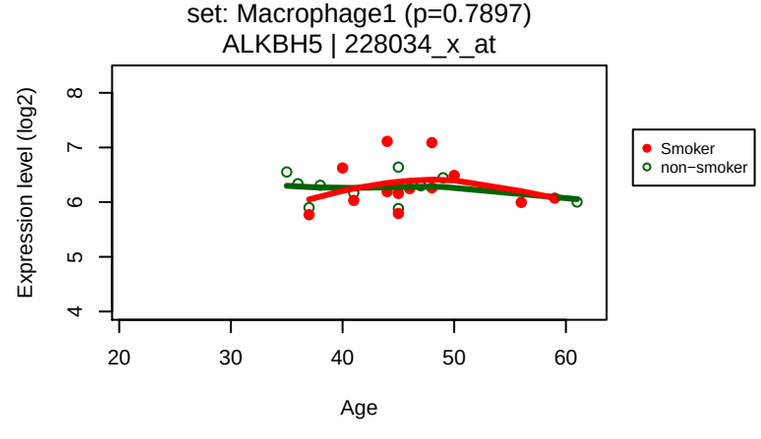

set: SmallAirway1 (p=0.0466)
ALKBH5 | 228034_x_at
Expression level (log2)
Age
Smoker
non−smoker

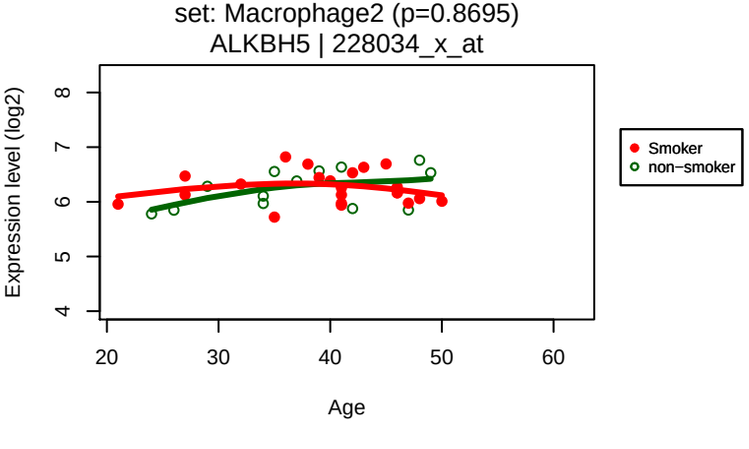

set: SmallAirway2 (p=0.5785)
ALKBH5 | 228034_x_at
Expression level (log2)
Age
Smoker
non−smoker

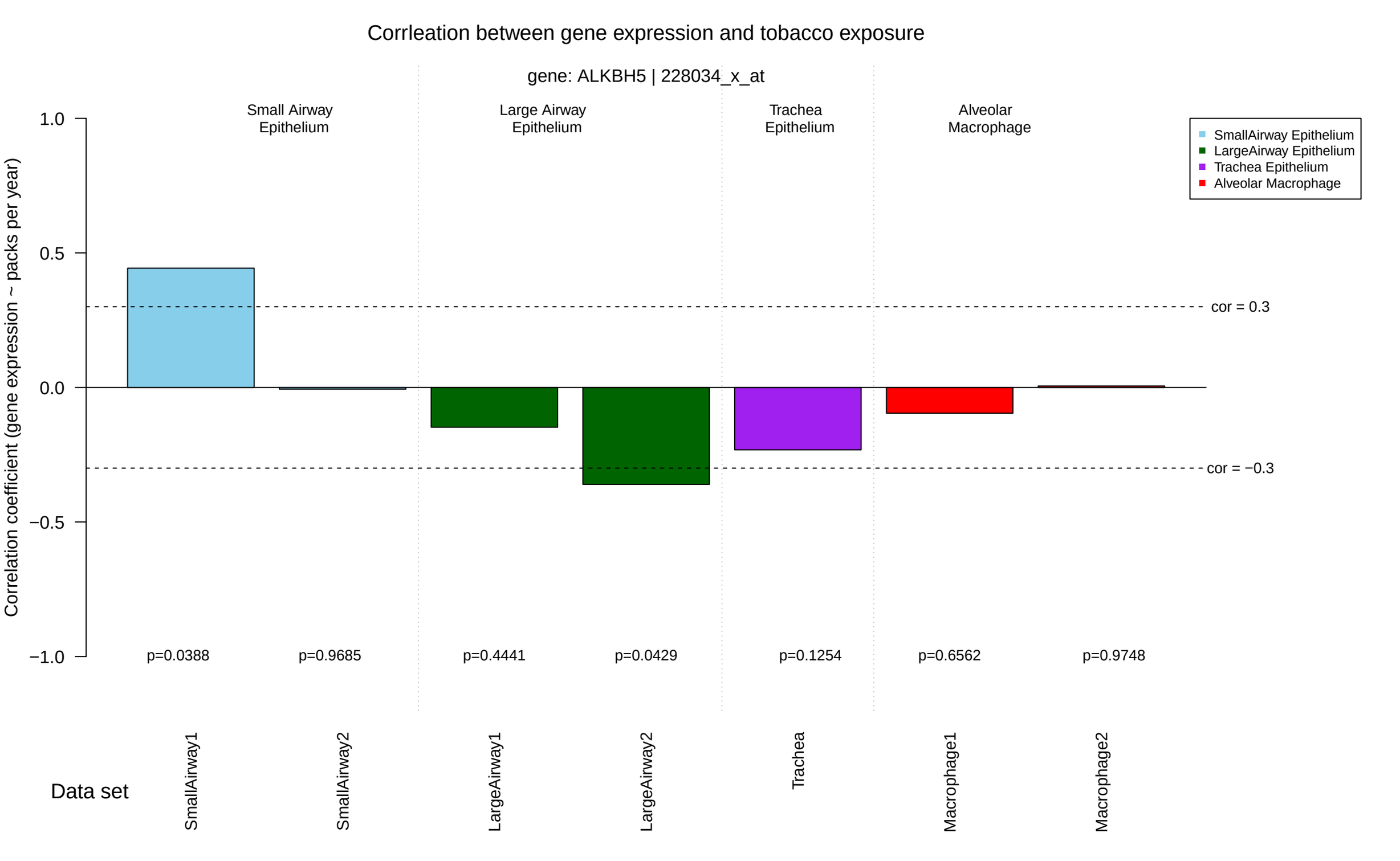

set: LargeAirway1 (p=0.8926)
ALKBH5 | 228034_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3519)
ALKBH5 | 228034_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0525)
ALKBH5 | 228034_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7897)
ALKBH5 | 228034_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8695)
ALKBH5 | 228034_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALKBH5 | 228034_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0388
p=0.9685
p=0.4441
p=0.0429
p=0.1254
p=0.6562
p=0.9748
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2